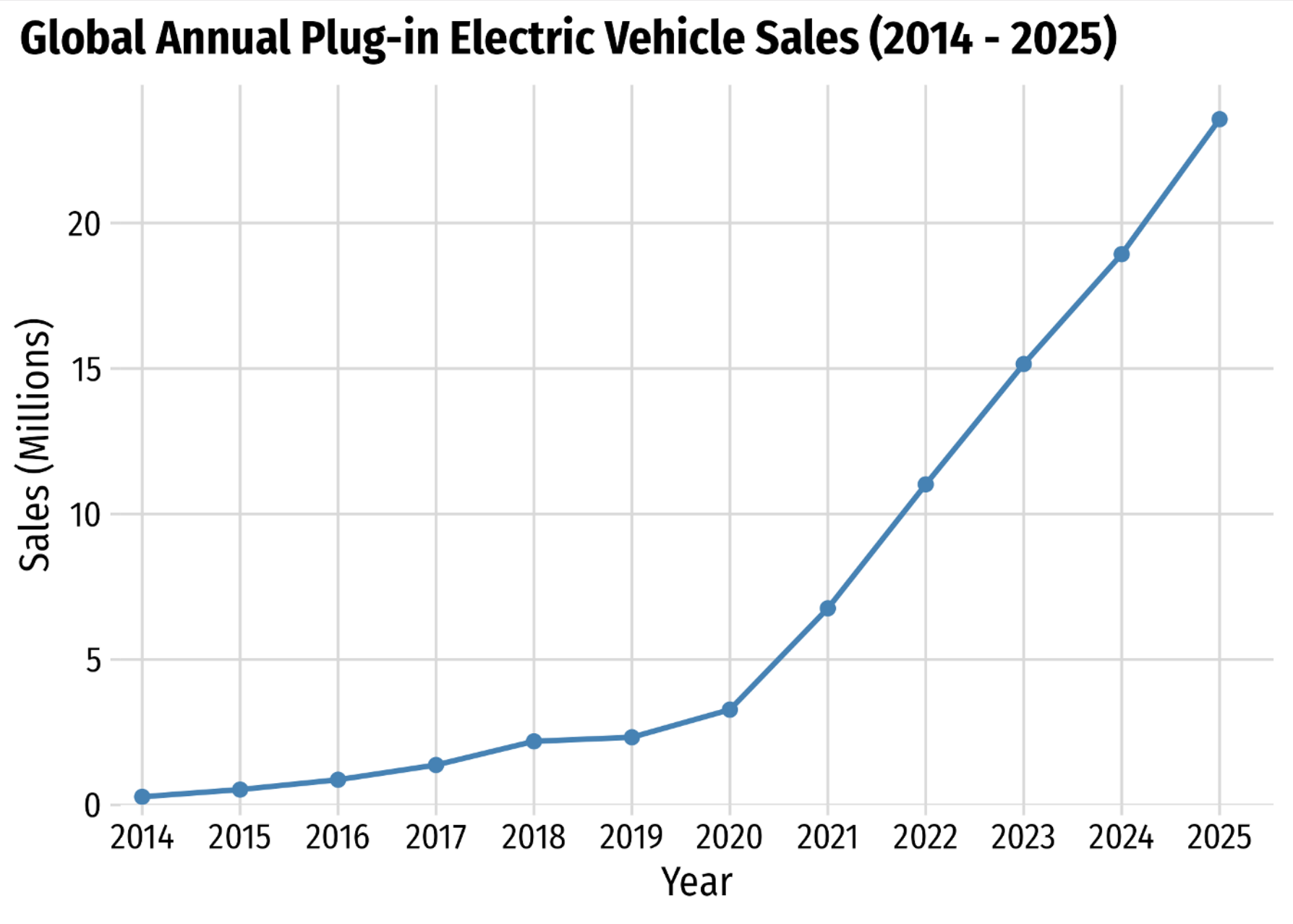
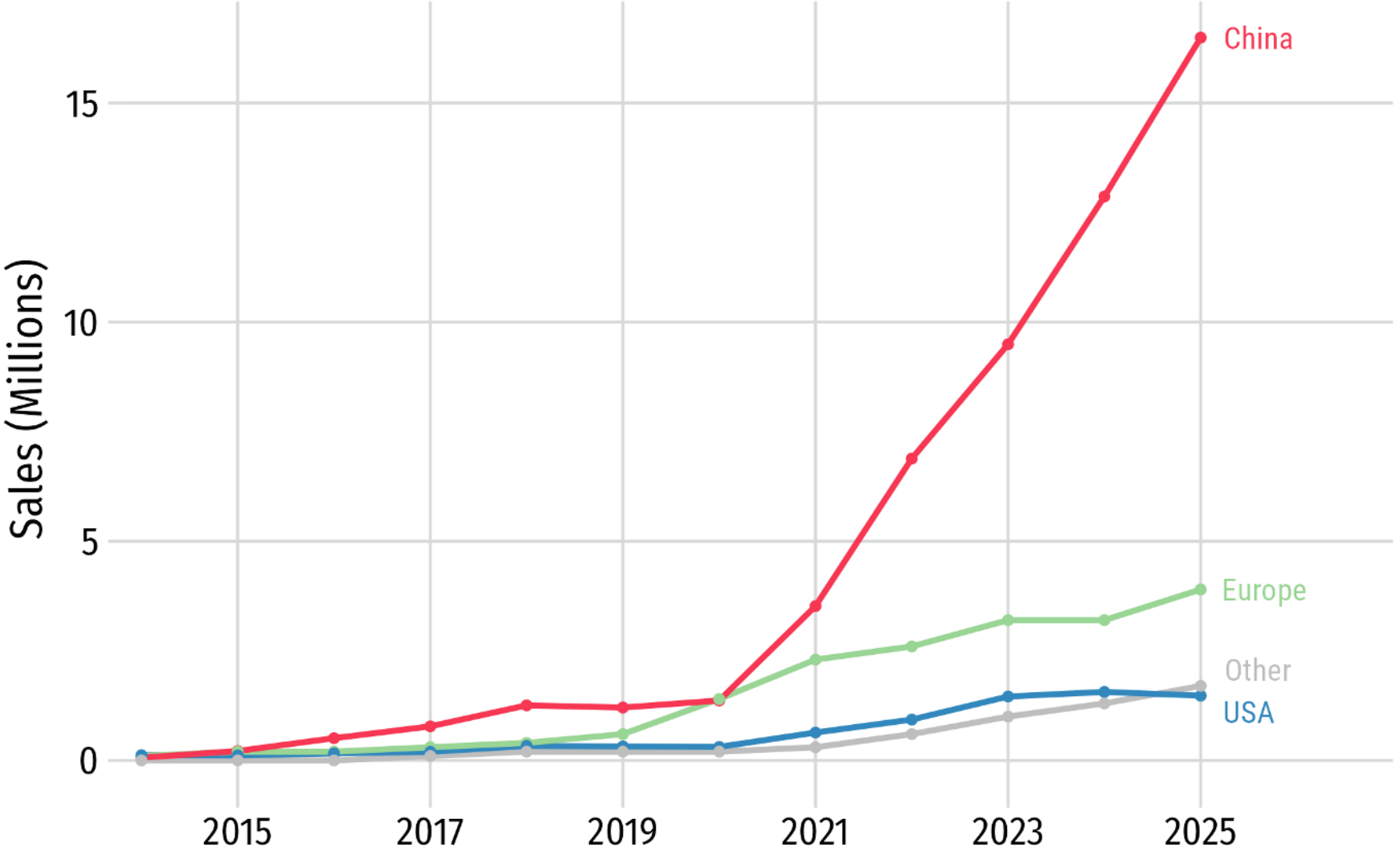
The American EV Transition: Looking Ahead While Lagging Behind (China)

John Paul Helveston
George Washington U.
April 22, 2026

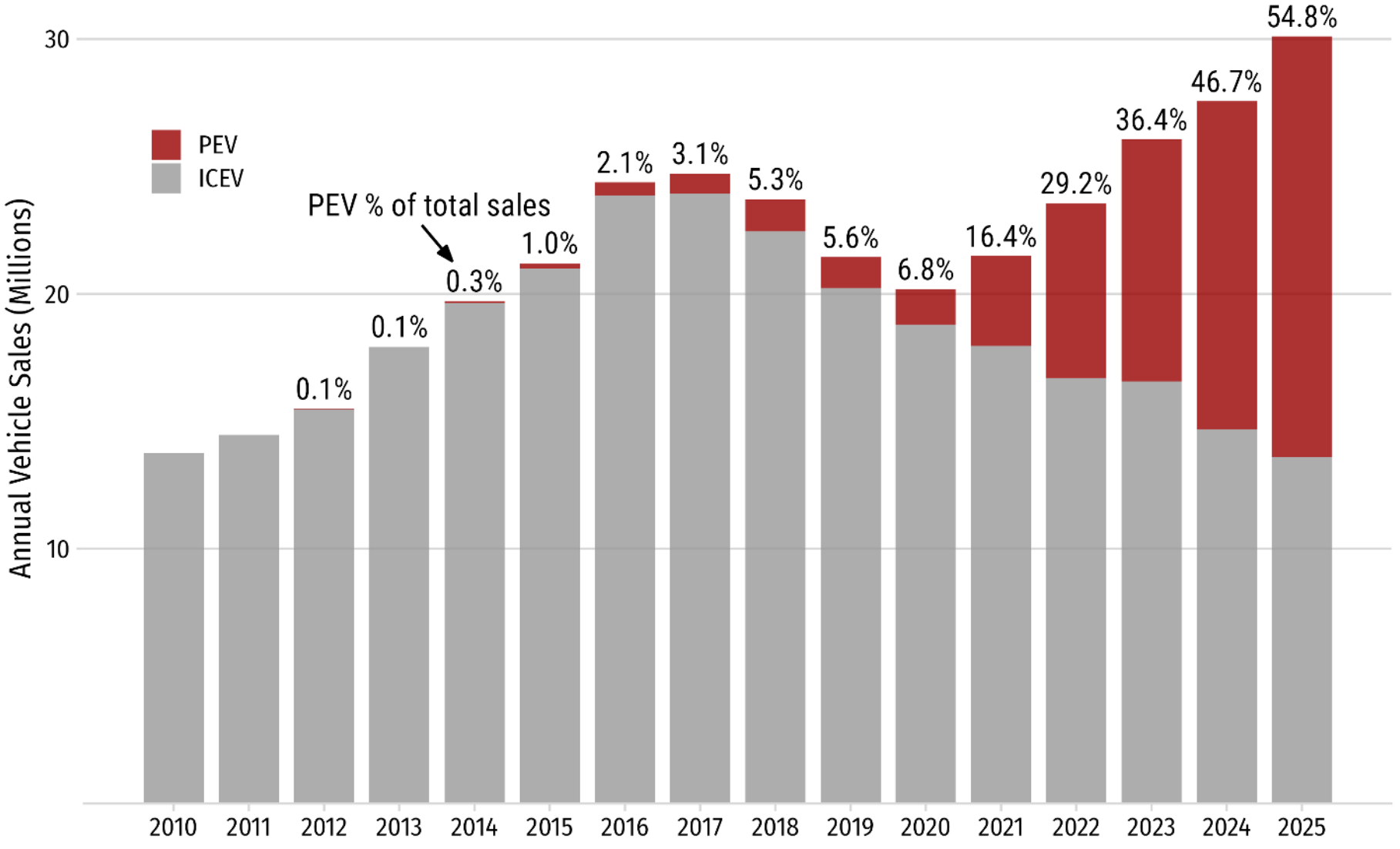

Global Annual Plug-in Electric Vehicle Sales (2014 - 2025)
Sales (Millions)
0
5
10
15
20
2014
2015
2016
2017
2018
2019
2020
2021
2022
2023
2024
2025
Year

China is driving the EV revolution



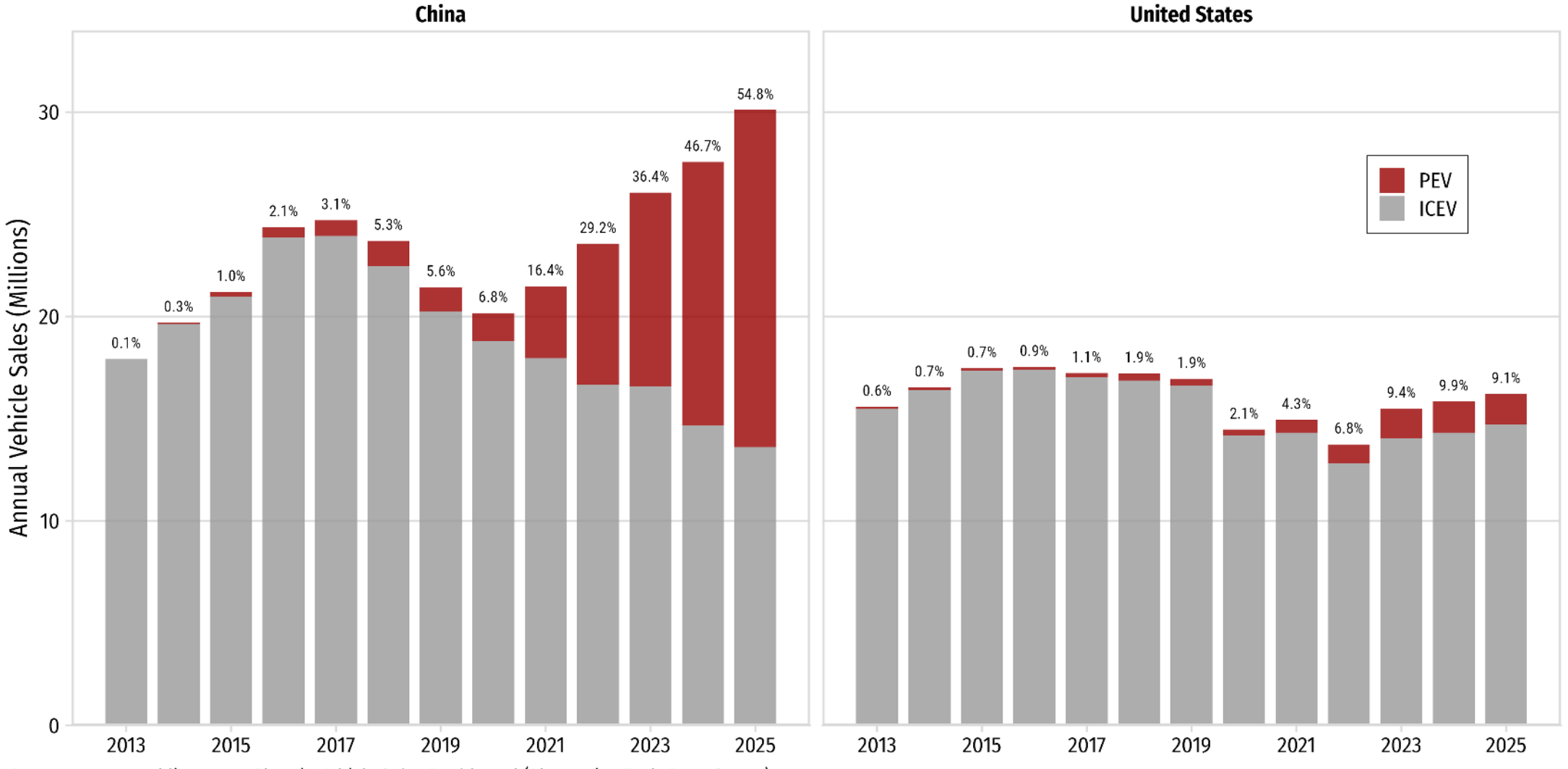
In China, PEV sales grow while ICEV sales slow

After peaking in 2017, internal combustion engine vehicle (ICEV) sales have declined for 8 straight years



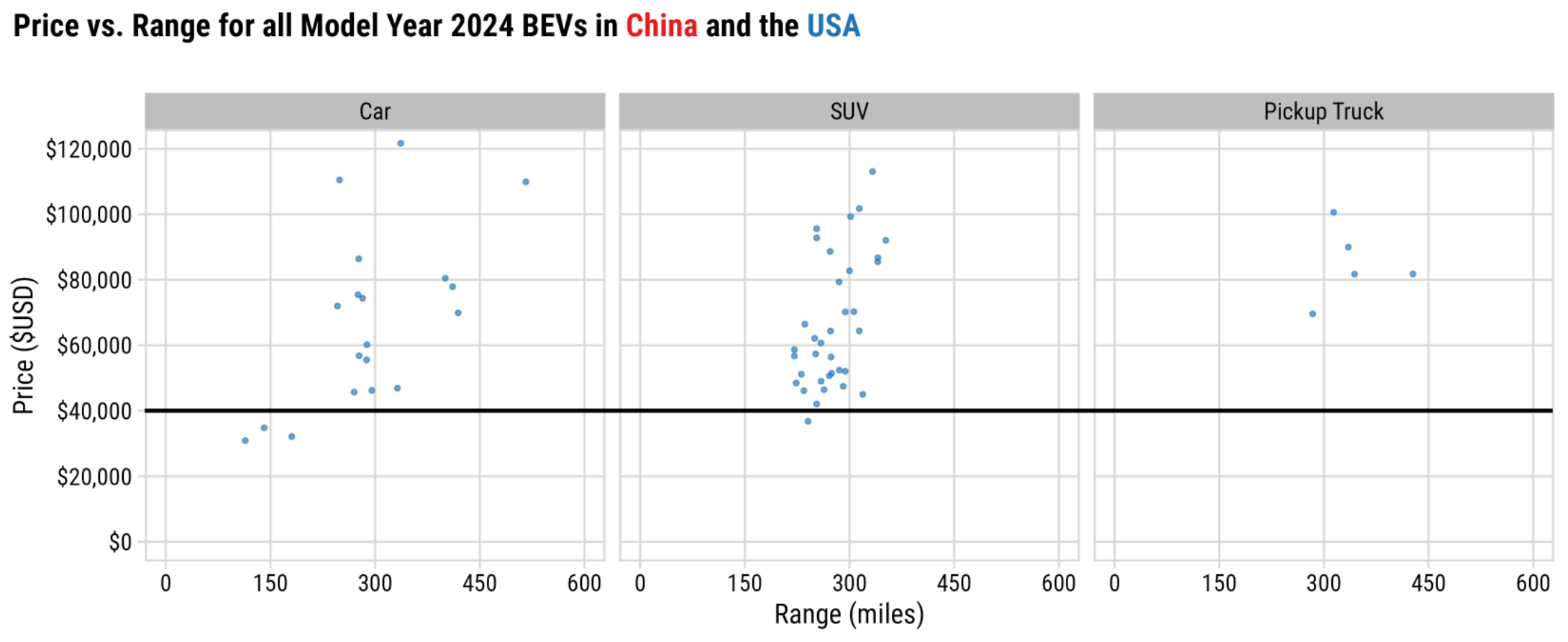
Data sources: OICA, marklines.com

PEV adoption in China and the United States



Data sources: marklines.com; Electric Vehicle Sales Dashboard (Alternative Fuels Data Center)

China offers more affordable BEVs across all range categories



Data scraped from autocango.com (China) and carsheet.io (USA)

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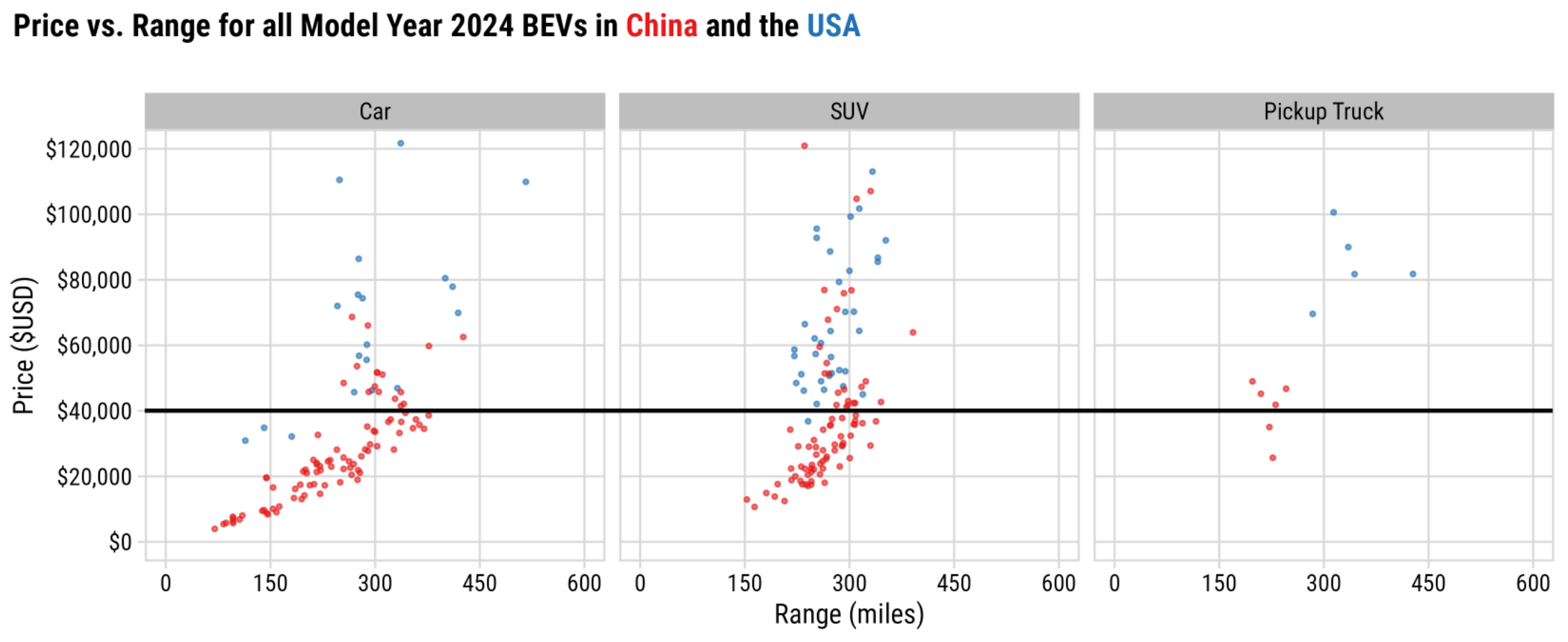

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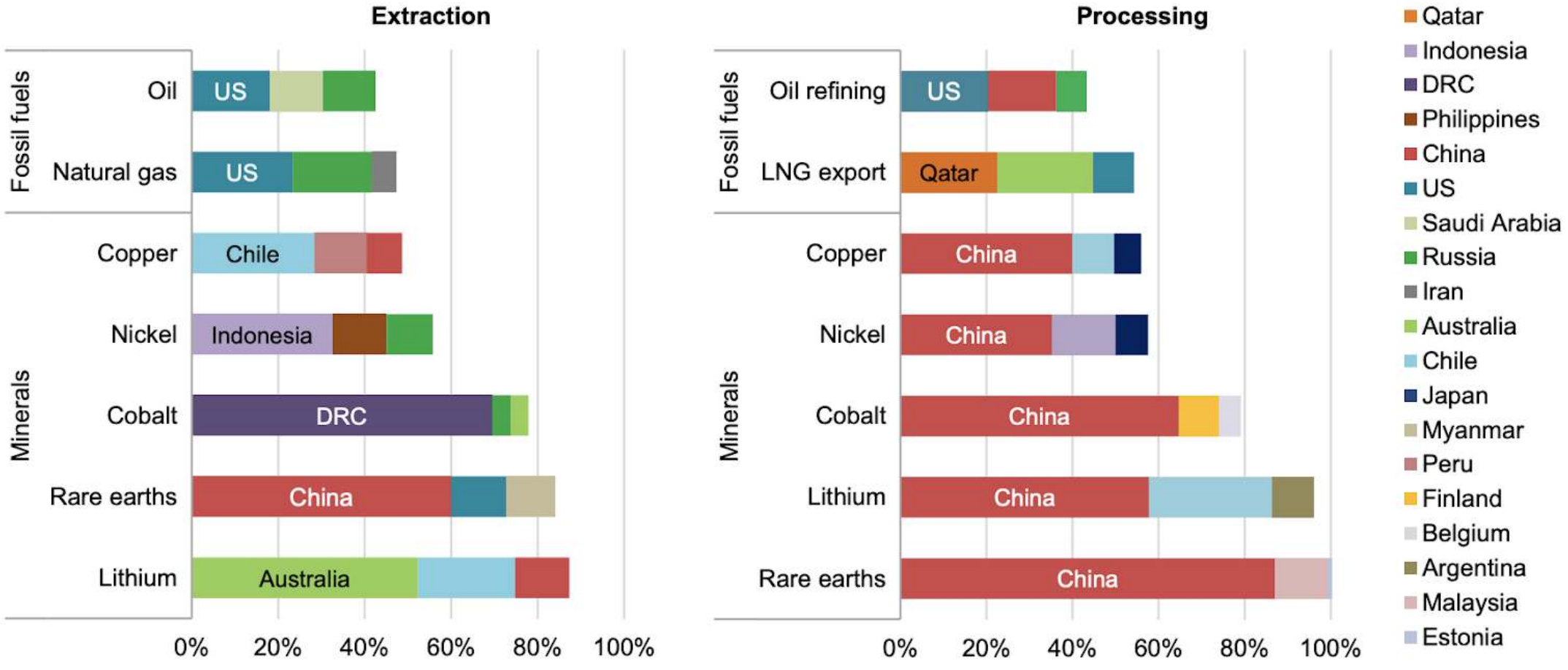
China dominates mineral processing supply chain



Source: https://thegraduatepress.org/2022/04/01/how-china-came-to-dominate-the-supply-chain-of-rare-earths-and-critical-minerals/

How did this happen?

Most common answer: it's the subsidies, right?!

Estimated $230B in subsidies (2009 - 2023)

Type of Support	2009-2017	2018	2019	2020	2021	2022	2023	Total
Rebate	37.8	4.3	3.3	3.5	7.4	9.2	0.0	**65.7**
Sales Tax Exemption	10.8	7.7	6.4	6.6	16.4	30.3	39.6	**117.7**
Infrastructure Subsidies	2.3	0.2	0.2	0.3	0.3	0.6	0.6	**4.5**
Research & Development	2.0	3.6	3.4	3.5	4.3	3.9	4.3	**25.0**
Government Procurement	7.8	1.6	1.4	2.9	1.7	1.8	0.8	**18.0**
Total	**60.7**	**17.4**	**14.8**	**16.8**	**30.1**	**45.8**	**45.3**	**230.9**
Spending as Share of Total Sales	42.4%	22.7%	23.3%	25.4%	18.3%	15.1%	11.4%	**18.8%**
Subsidy per Vehicle (US$)	-	13,860	12,311	12,294	8,538	6,656	4,764	-

CSIS | TRUSTEE CHAIR IN CHINESE BUSINESS & ECONOMICS

https://www.csis.org/blogs/trustee-china-hand/chinese-ev-dilemma-subsidized-yet-striking

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50% sales tax exemptions
30% direct rebates

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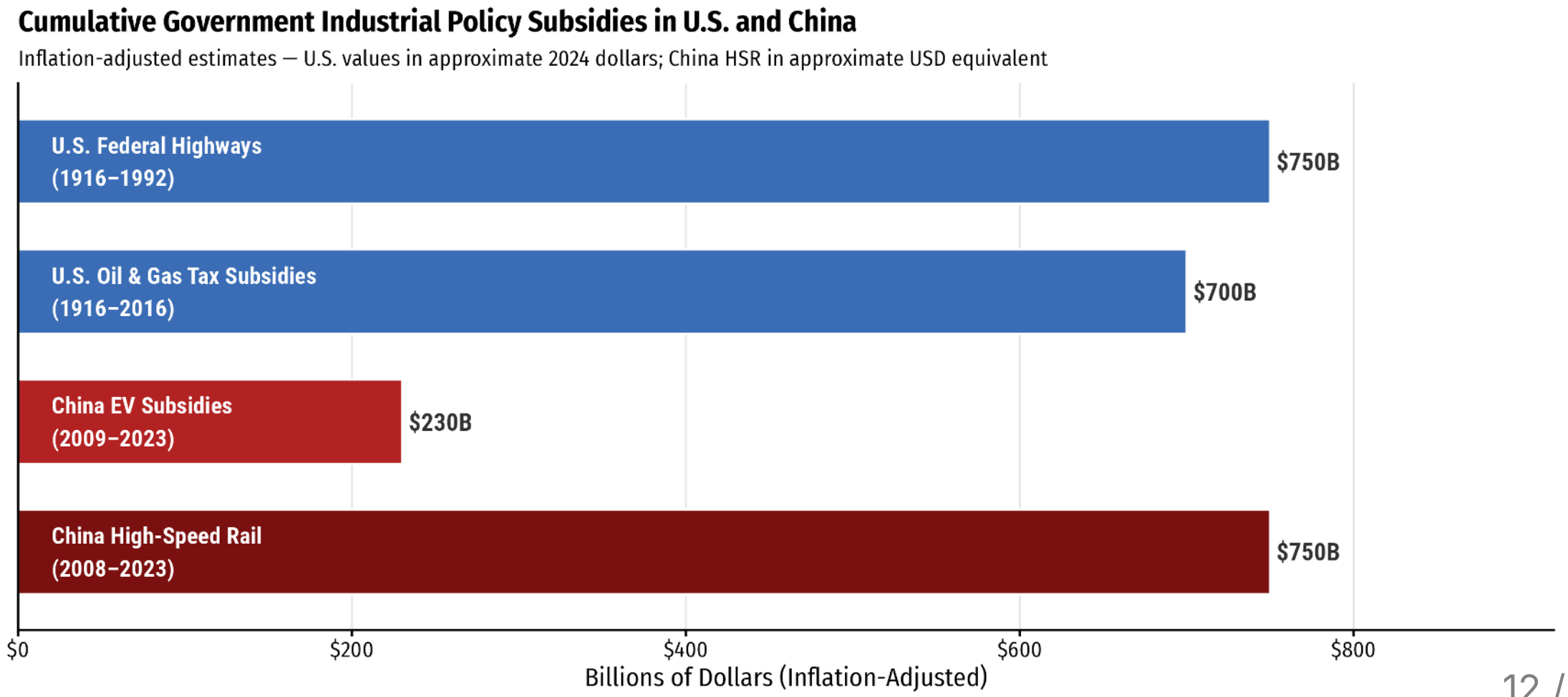
China's EV subsidies are on par with typical subsidies needed to start an industry



Institutions Matter

诸侯
经济
Local
Protectionism

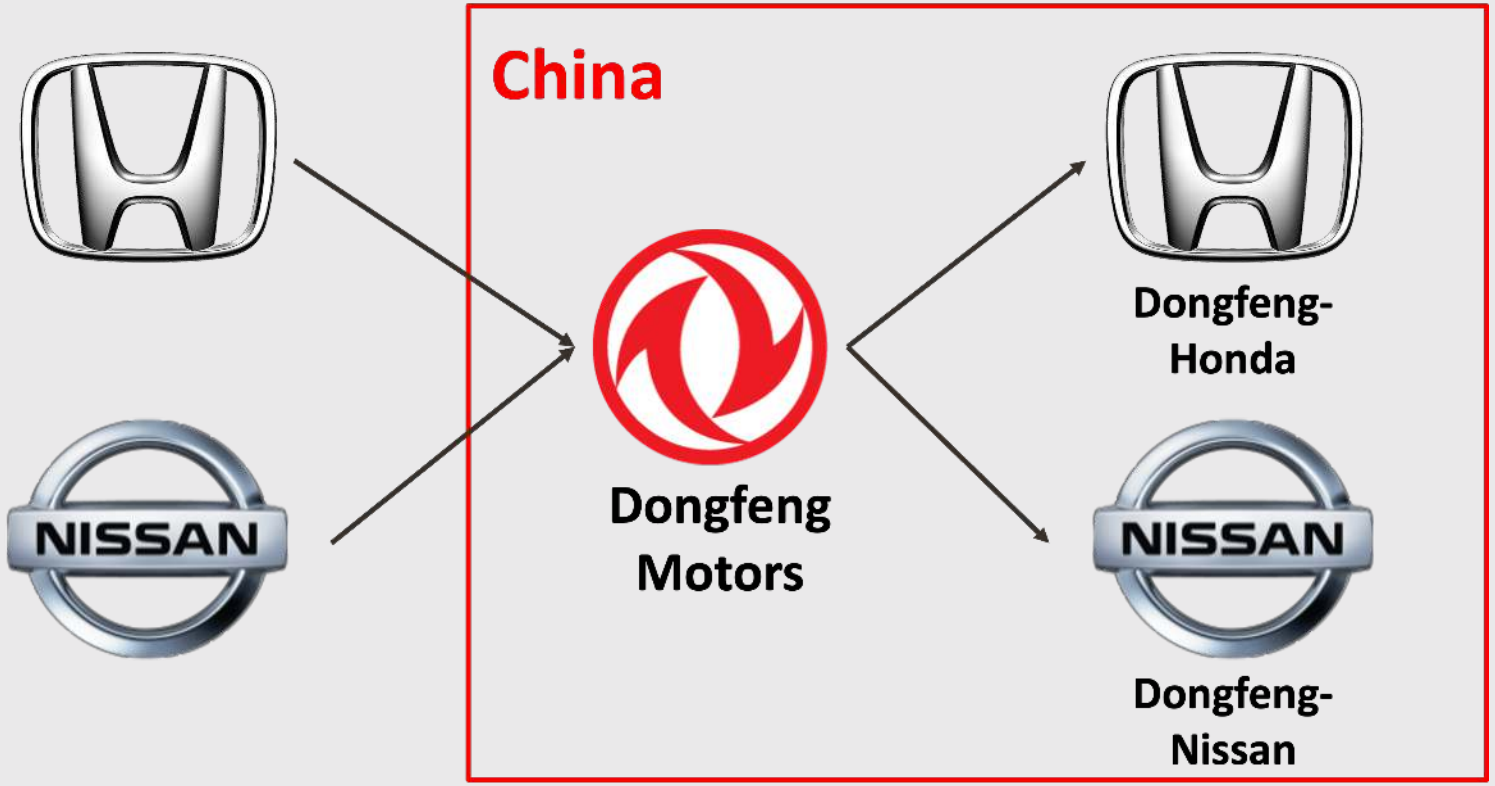
内卷
Involution
& “996”
诸侯
经济
Local
Protectionism

R&D focus on production innovation

Why are *Chinese* firms making the best EVs?
(and not multinationals)

The Chinese Joint Venture System

1980s: 以市场换技术 = “Exchange market for technology”



“这就像吸食鸦片一样，一旦你沾染上了就永远也无法戒掉。”

何光远, 中国前机械工业部部长

“It's like opium. Once you've had it you will be addicted forever.”

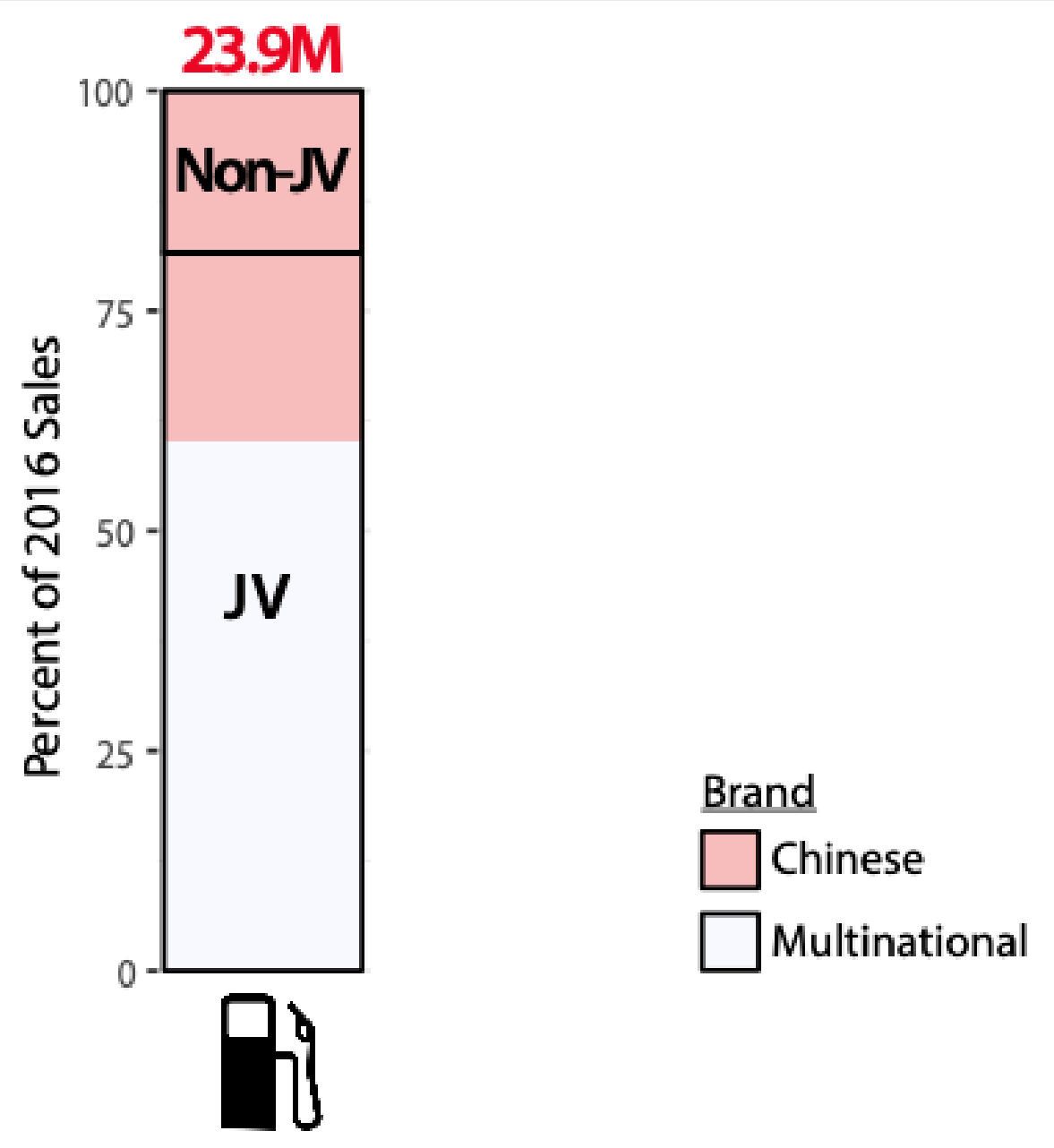
Guangyuan He, Former Minister of Machinery and Industry (Reuters, 2012)

Who's going to make the EVs?

❌ Multinationals don't want to share cutting-edge technology

❌ Chinese JV partners lack incentives to innovate

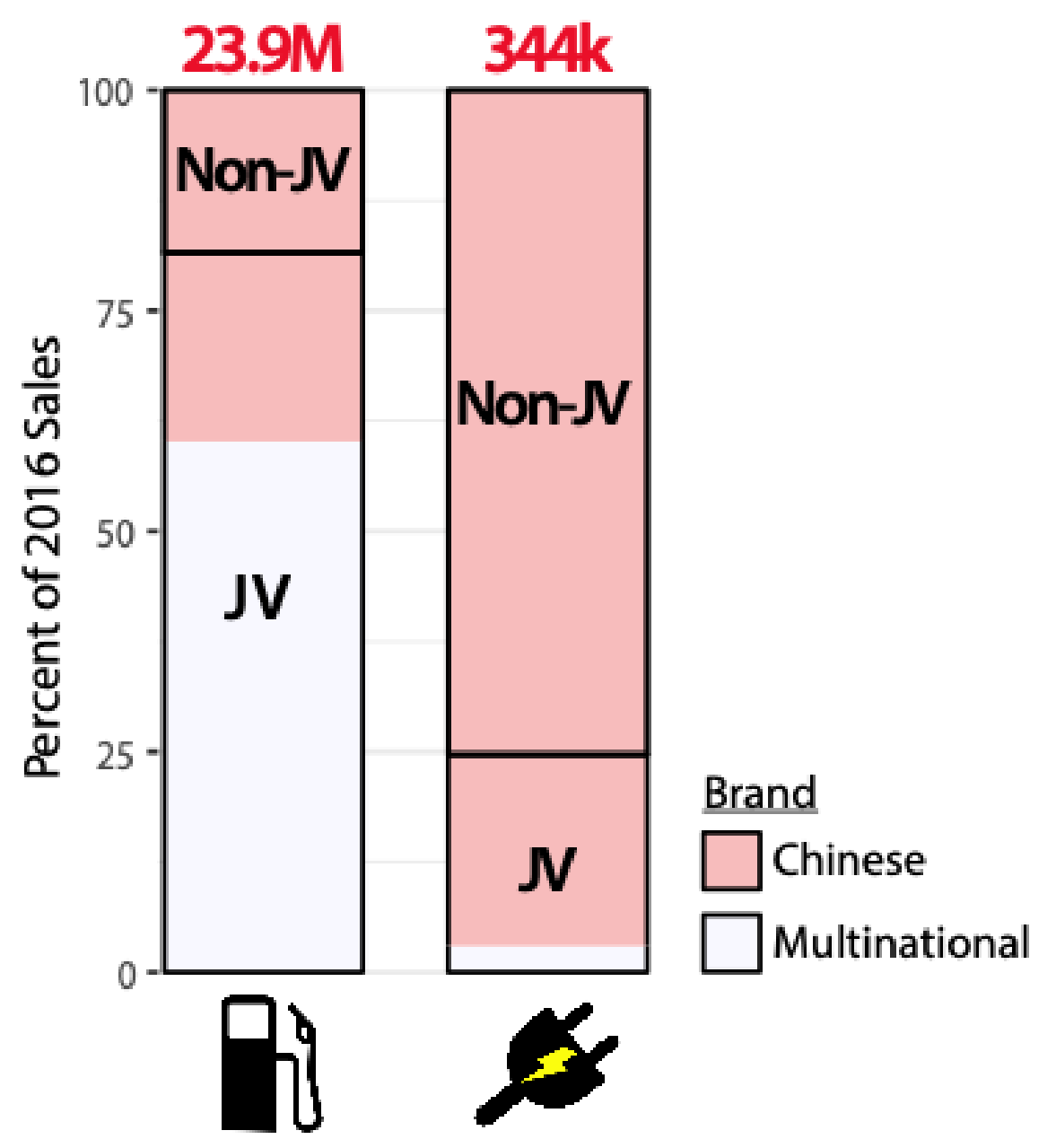
✅ Indigenous non-JV firms



Multinational OEMs dominate traditional auto sector through large JVs

Indigenous non-JV firms dominate emerging EV sector

Helveston, J. P., Y. Wang, V. Karplus, E. Fuchs (2018) "Institutional Complementarities: The Origins of Experimentation in China's Plug-in Electric Vehicle Industry." *Research Policy*. 48(1), pg. 206-222



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EV Challenges in the US

Decades of Policy Whiplash

Obama (2009–2016)

- $7,500 EV tax credit
- Goal: 1M EVs by 2015
- DOE battery R&D investment

Trump I (2017–2020)

- Proposed eliminating EV tax credits
- Rolled back fuel economy standards
- Withdrew from Paris Agreement

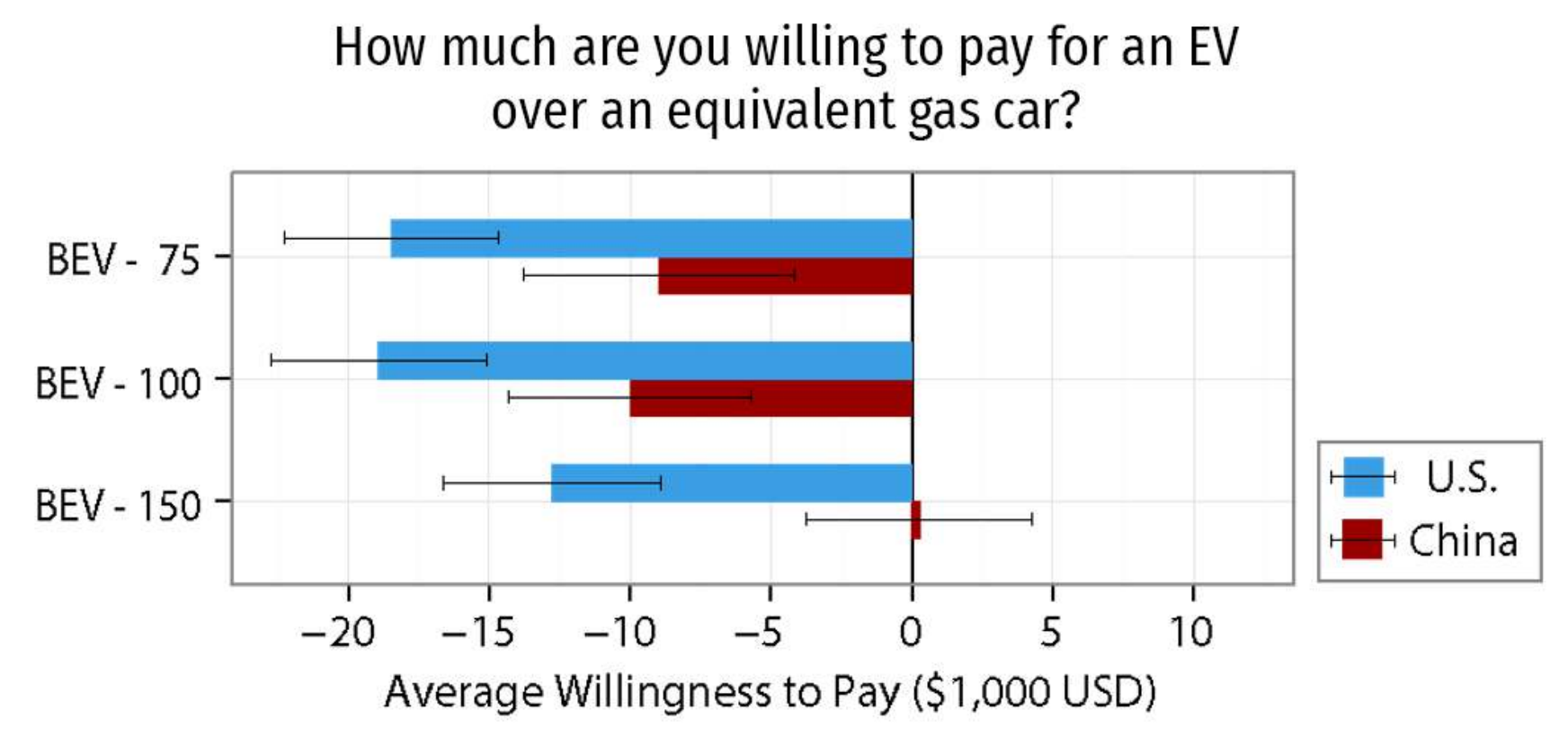
Biden (2021–2024)

- IRA: $369B in clean energy investment
- Expanded EV credits, domestic content rules
- 100% EV federal fleet goal

Trump II (2025–Now)

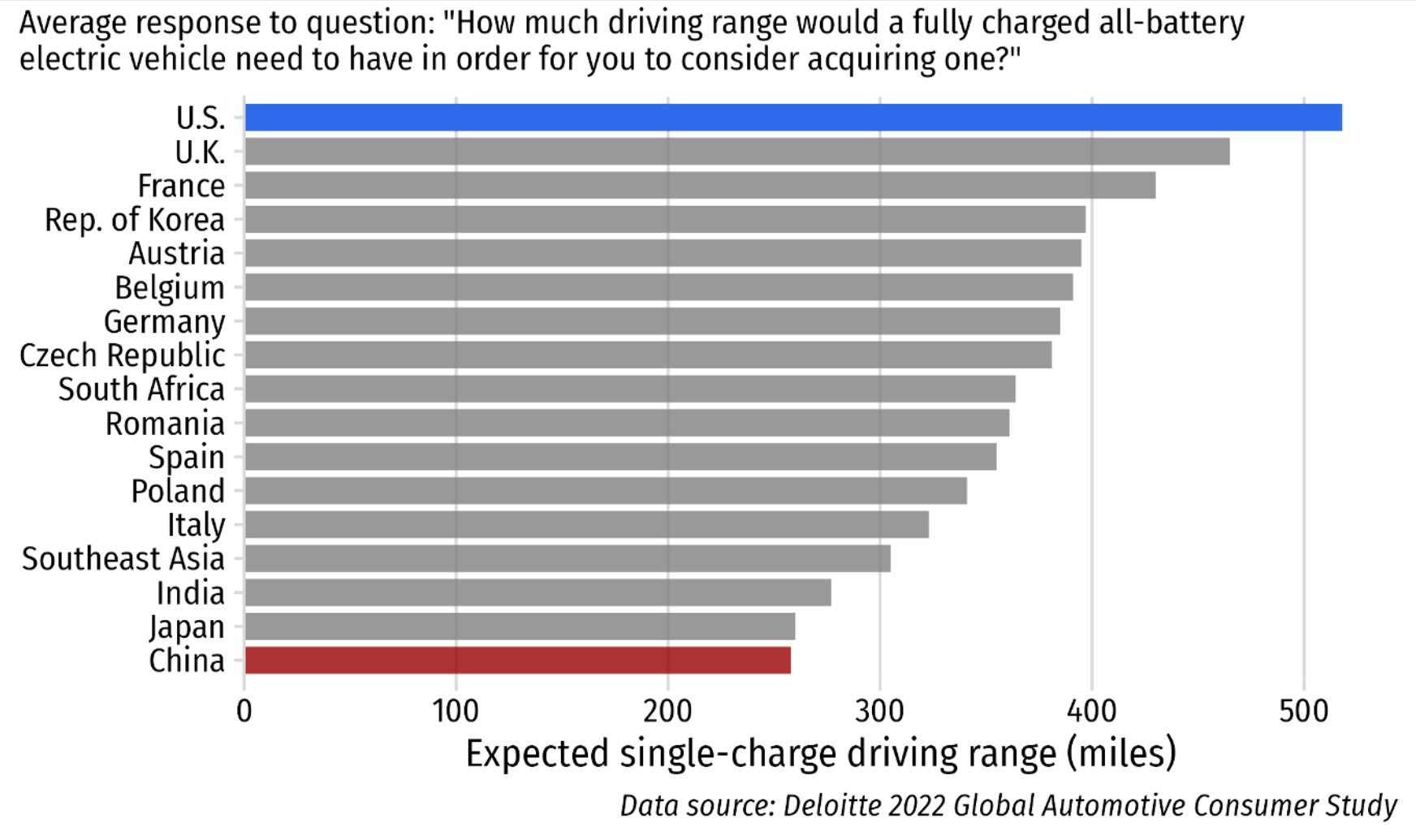
- Gutted IRA
- Rolled back fuel economy standards

US buyers are more opposed to EVs than Chinese buyers



Helveston et al. (2015) "Will subsidies drive electric vehicle adoption? Measuring consumer preferences in the U.S. and China" *Transportation Research Part A: Policy and Practice*. 73, 96–112. DOI: 10.1016/j.tra.2015.01.002

US buyers demand far more driving range than Chinese buyers



Top 4 Selling Vehicles in **China**

Xiaomi SU7
$30,171 - $41,909

BYD Song
$24,721 - $38,555

Geely Xingyuan
$ 9,615 - $13,667

Tesla Model Y
$36,822 - $43,809

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Top 4 Selling Vehicles in **USA**

Toyota RAV4
$29,250 - $38,955

Honda CRV
$30,100 - $41,100

Ford F-150
$44,095 - $79,005

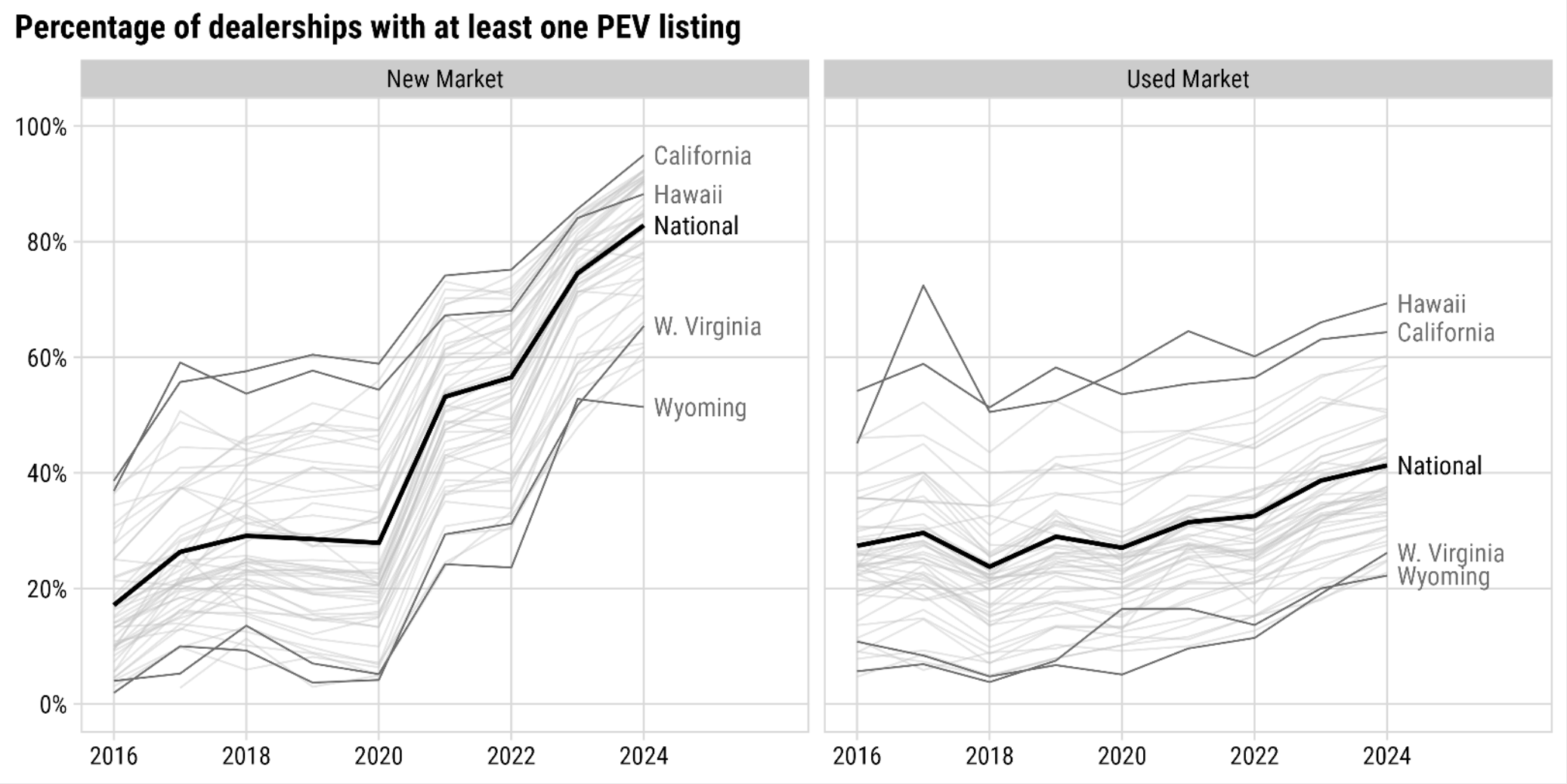
Chevrolet Silverado
$42,700 - $70,000

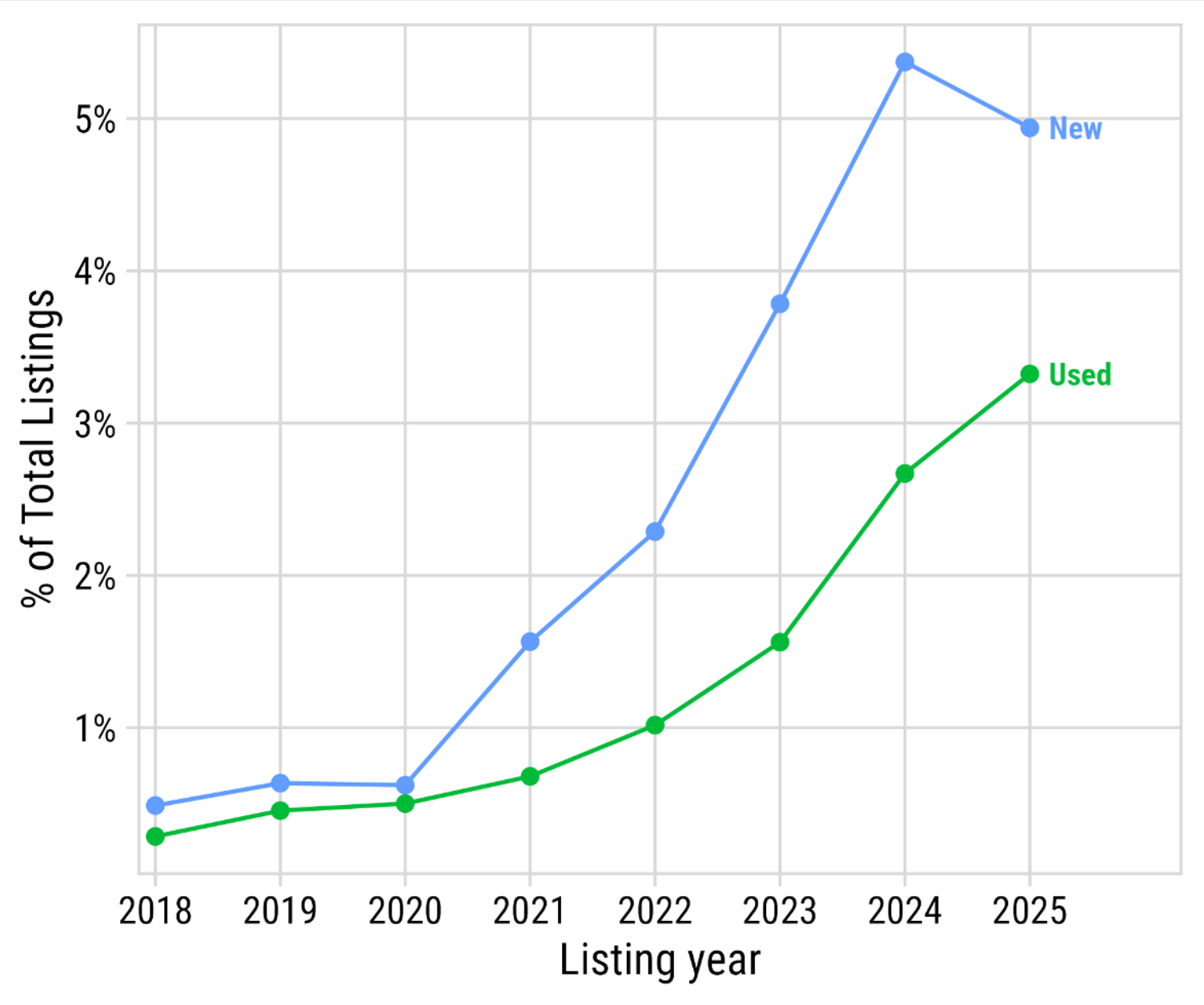
Using **vehicle listings** to quantify market development

128M vehicle listings from 60k dealerships
(2016 - 2025, inclusive)

How many dealerships are carrying EVs?

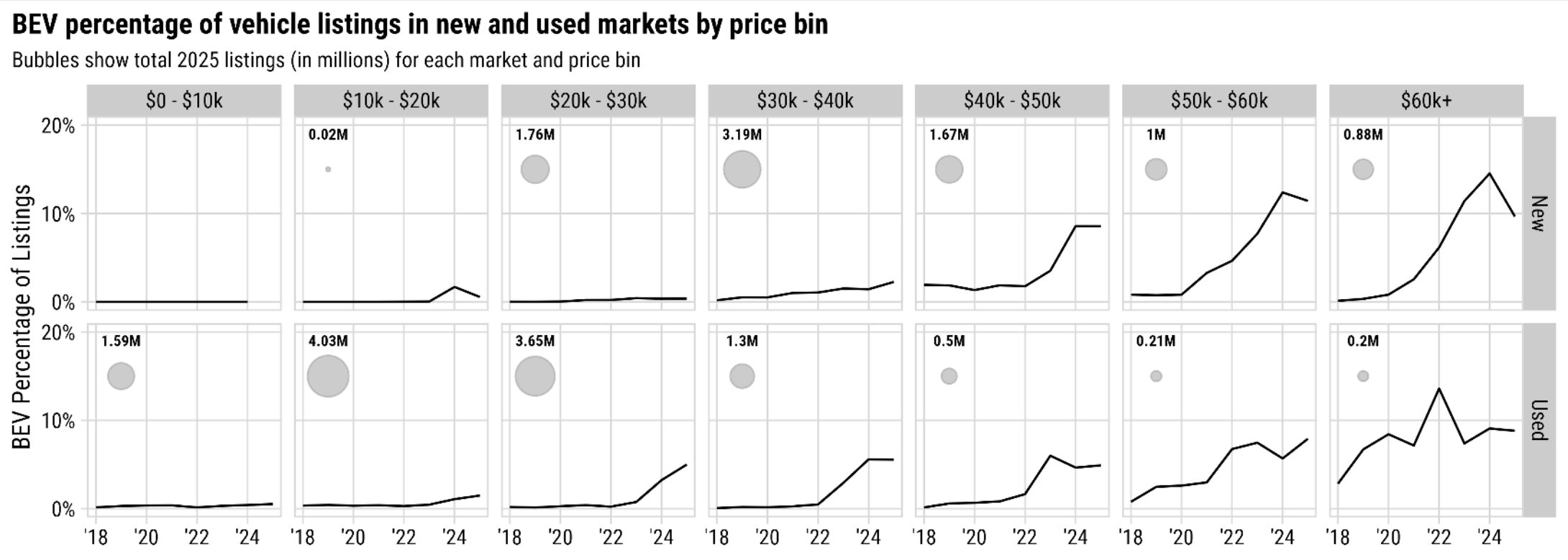
4/5 dealers have *new* BEV; **2/5** dealers have *used* BEV





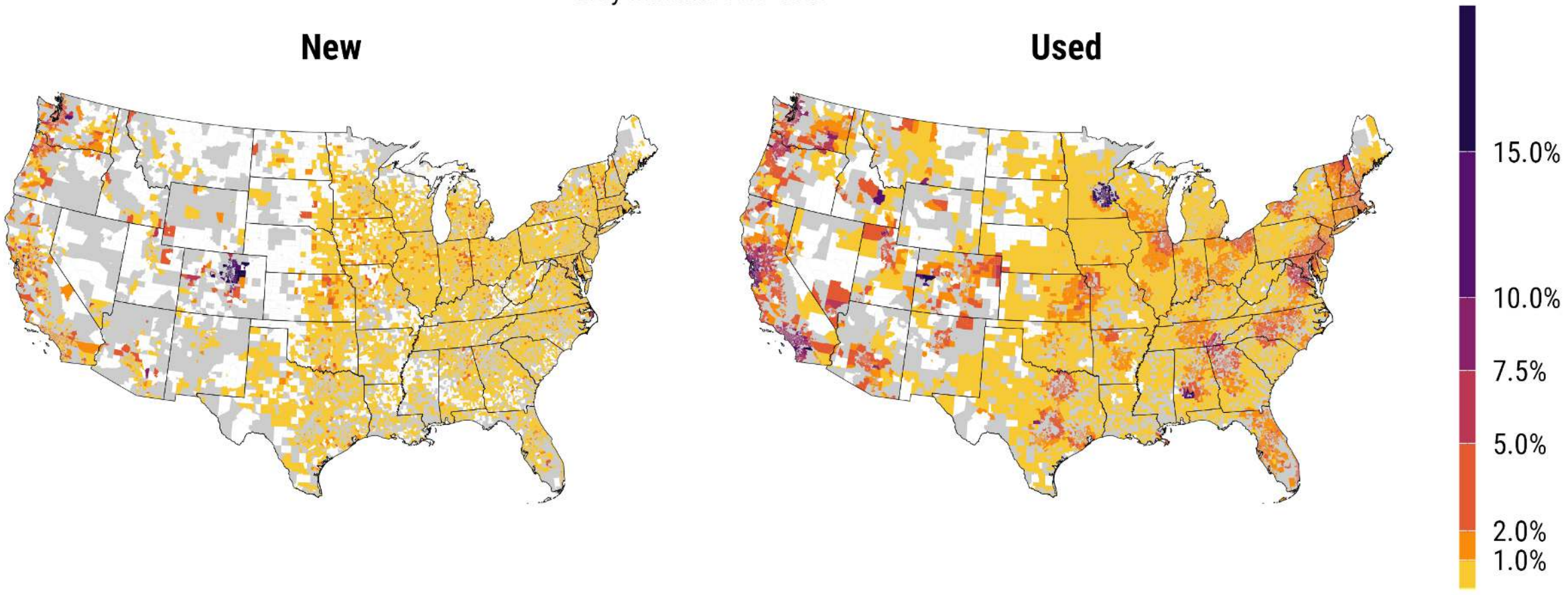
Most dealers carry very few EVs

Majority of BEV listings in high-price segments



BEV % of available vehicles in local market within budget

White indicates 0% BEVs
Gray indicates no data



How should the US respond to Chinese EVs?

Bipartisan Response (So Far): **Ban the Chinese tech**

Lutnick Shuts Down Talk of Any Chinese Investment in US Autos

Howard Lutnick at the Semafor World Economy Summit in Washington on April 17. *Photographer: Aaron Schwartz/Bloomberg*

By Colum Murphy

April 17, 2026 at 5:50 PM EDT

Block imports: *Steep tariffs on imported Chinese EVs, batteries (First Biden, continued w/Trump)*

Block investment: *Stricter Foreign Entities of Concern (FEOC) rules*

Wrong lesson

- China cheated its way to the top with subsidies
- Chinese EVs are a security threat
- Chinese EVs will destroy the US auto sector

→ **Exclude them**

Right lesson

- China built real competitive advantages in manufacturing and supply chains
- Chinese EV tech is winning global markets

→ **Engage strategically**

The Tesla Example

Targeted compliance, not exclusion:

- Store vehicle data **within China**
- Restrict operations near **sensitive areas**
- Use only **government-approved** mapping systems (Baidu)

The European Model

Tariffs to decrease imports while welcoming investment

- Access to cutting-edge battery tech
- Create local jobs
- Increase supply chain resilience

CATL

$7.6B battery plant in Debrecen

100 GWh/year for Mercedes-Benz, BMW, Volkswagen

BYD

First European EV manufacturing facility

Producing BEVs and PHEVs for the European market

US Opportunities

Technology Licensing Agreements

Ford-CATL: Licensing battery technology in a Michigan plant

Challenge: CATL recently added to DOD's list of "Chinese military companies"

Chinese FDI into U.S.

Gotion batteries: Multi-billion dollar investments in Illinois and Michigan

Challenge: Uncertainty around Foreign Entities of Concern (FEOC) status

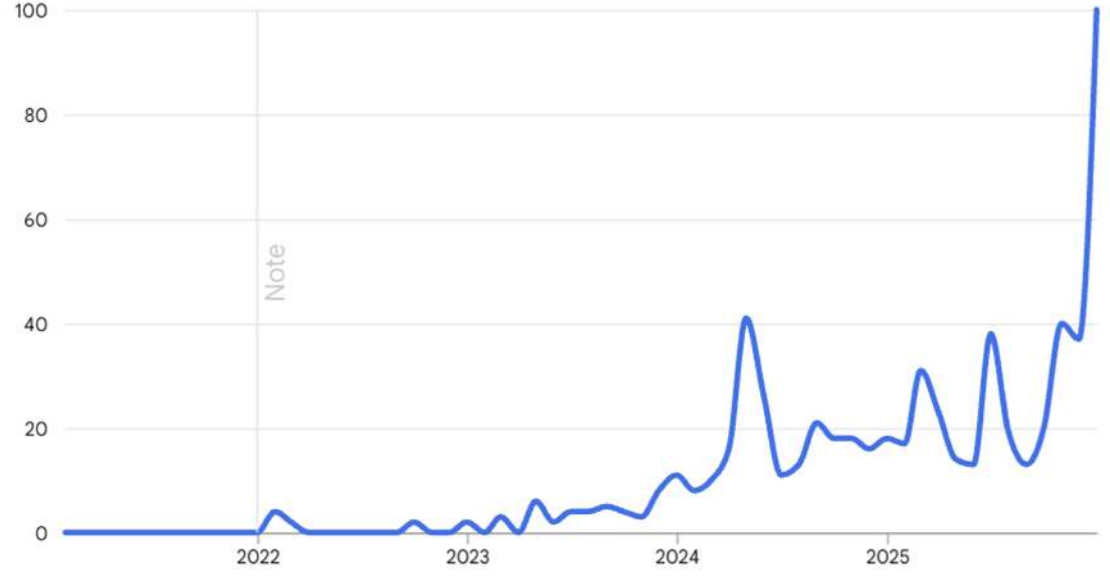
Consumers aren't waiting for Washington

Chinese EVs
Search term

United States | Past 5 years | Web Search

Interest over time

United States · Past 5 years



Rising queries

United States · Past 5 years

	Query	Search interest	Change
1	chinese evs canada		BREAKOUT
2	canada chinese evs		BREAKOUT
3	byd		BREAKOUT
4	byd evs		BREAKOUT
5	best chinese evs		BREAKOUT
6	chinese ev cars		BREAKOUT
7	canada tariffs chinese evs		BREAKOUT
8	byd ev		BREAKOUT
9	byd car		BREAKOUT
10	nio		BREAKOUT

Public Opinion

CHATHAM HOUSE

Topics Regions Events Publications Become a

Home > Topics/Regions >

Almost 70% of people globally are comfortable with their government buying Chinese-made clean technologies, new survey finds

Bloomberg

Live TV Markets Economics Industries Tech Politics Businessweek Opinion Video More

Electric Vehicles: Gas Price Tipping Point | Hybrids' New Popularity | Cutting Oil Demand | E-Bikes Hit a Bump | China's EV Battery

Industries
Transportation

Canadians Say They'll Buy Cheaper Chinese EVs as Tariffs Drop

Exclusive: Trump pollster finds MAGA voters like solar energy

Ben Geman

Add Axios on Google

COX AUTOMOTIVE

Brands What We Offer Insigh

1. Market polarization defines opportunity — and risk

Consumer sentiment toward Chinese auto brands is sharply divided. Younger, EV-oriented shoppers show meaningful openness, while older and domestic-loyal buyers remain resistant.

Consideration of Chinese Brands in the U.S.

38% extremely/very likely

39% not very/not at all likely

69% of Gen Z more likely

Ford's CEO Doesn't Want To Stop Driving China's Apple Car

Ford CEO Jim Farley has been driving an imported Xiaomi SU7 for months. He says it's fantastic.

The U.S. risks becoming an **"island of tailpipes"**

Strategic Responses

🇨🇦🇲🇽 **North American Integration**: Build sufficient scale through Canada and Mexico partnerships

📋 **Transparent Security Standards**: Develop compliance frameworks all connected vehicles must meet — *regardless of origin*

🔋 **Battery Manufacturing Focus**: Prioritize domestic production as platform for commercializing next-gen U.S. technologies

🤝 **Strategic Chinese Partnerships**: Pursue licensing agreements and FDI with targeted safeguards — not wholesale restrictions

📈 **Sustained Policy Support**: Restore and maintain IRA incentives long enough for investments to mature

Thanks!

Slides available at jhelvy.com/slides

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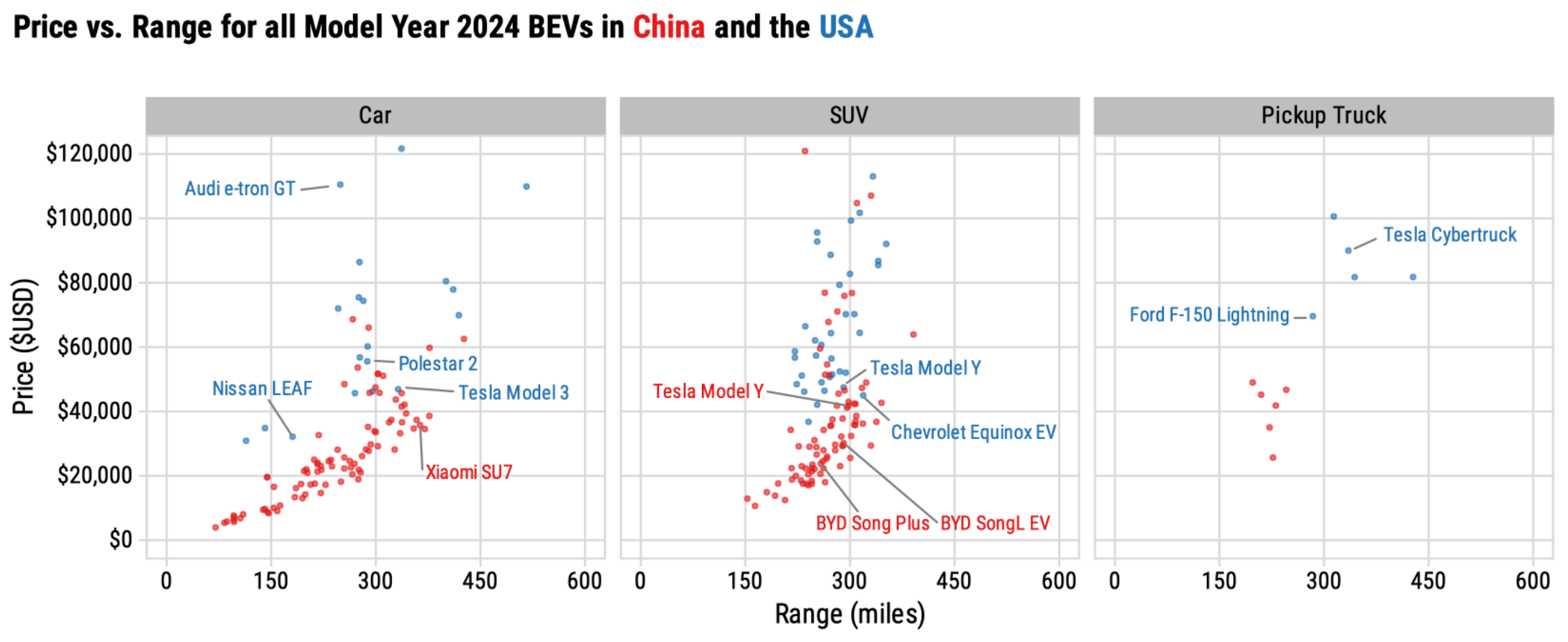
@jhelvy.bsky.social

Extra Slides

China offers more affordable BEVs across all range categories



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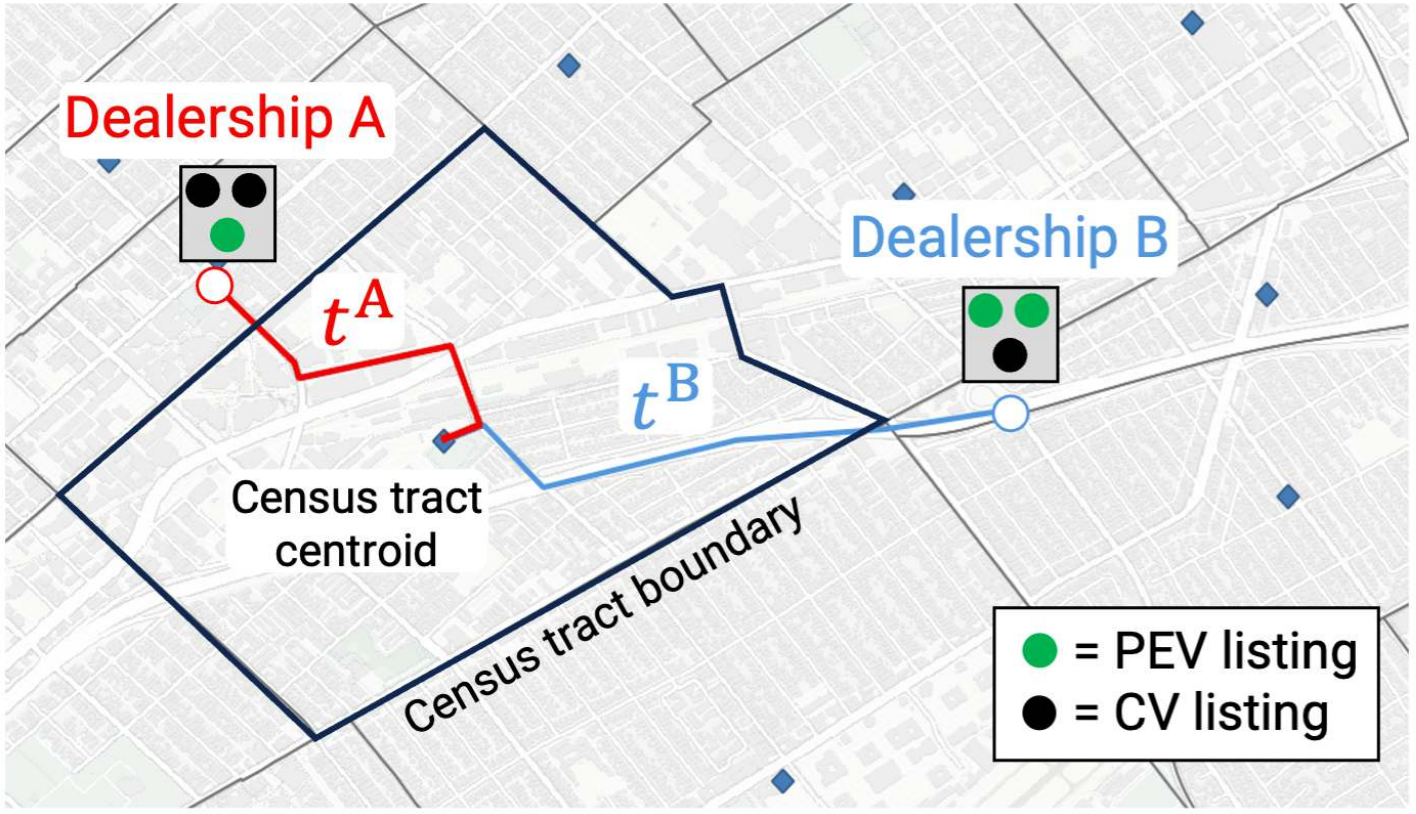
How hard is it to get to a PEV dealer?

Vehicle accessibility metric:
Road travel time from census tract centroid to nearest dealership with a target vehicle



Times to first 3 ● (PEVs)

$t_1^{PEV} = t^A$

$t_2^{PEV} = t^B$

$t_3^{PEV} = t^B$

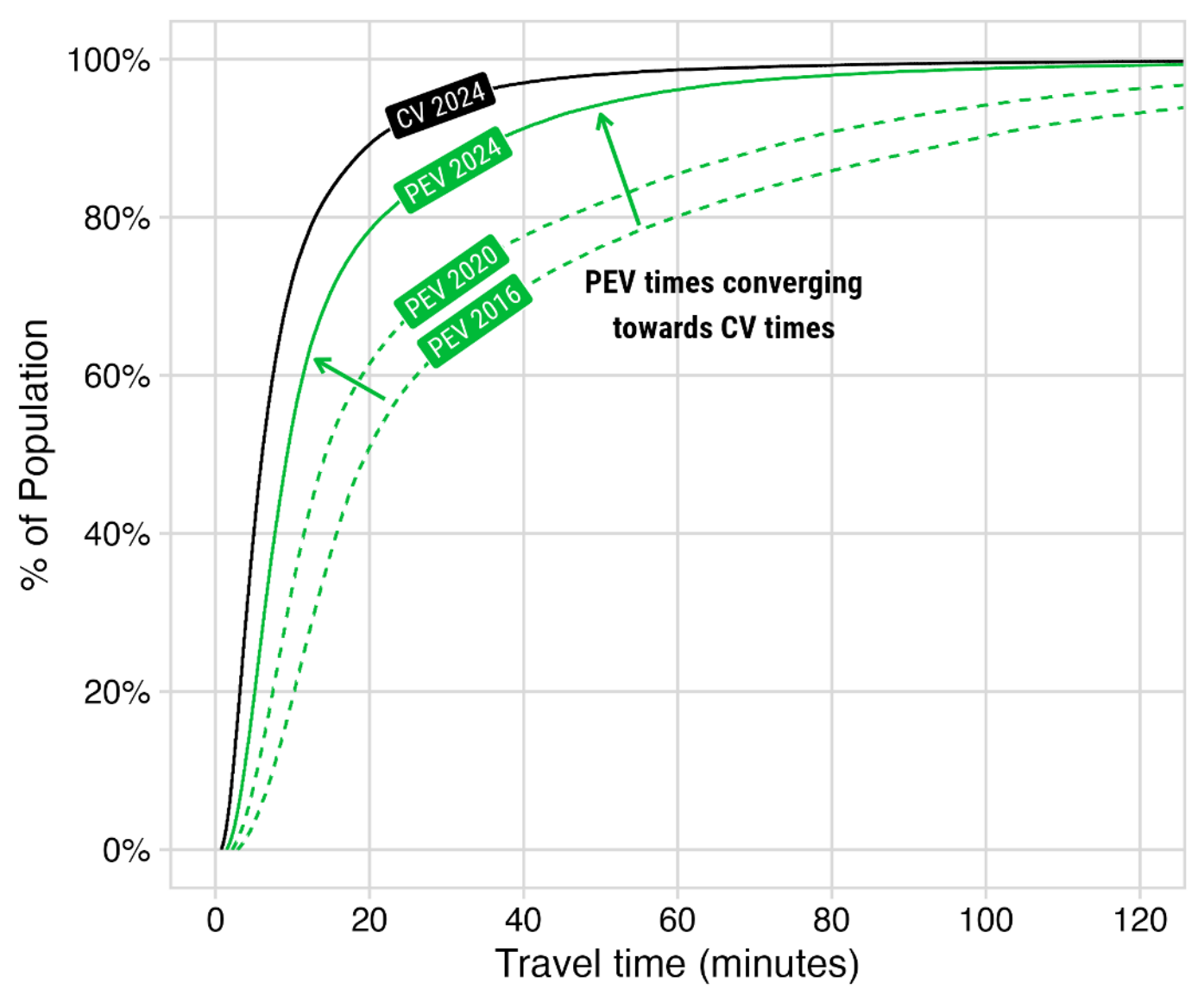
Times to first 3 ● (CVs)

$t_1^{CV} = t^A$

$t_2^{CV} = t^A$

$t_3^{CV} = t^B$

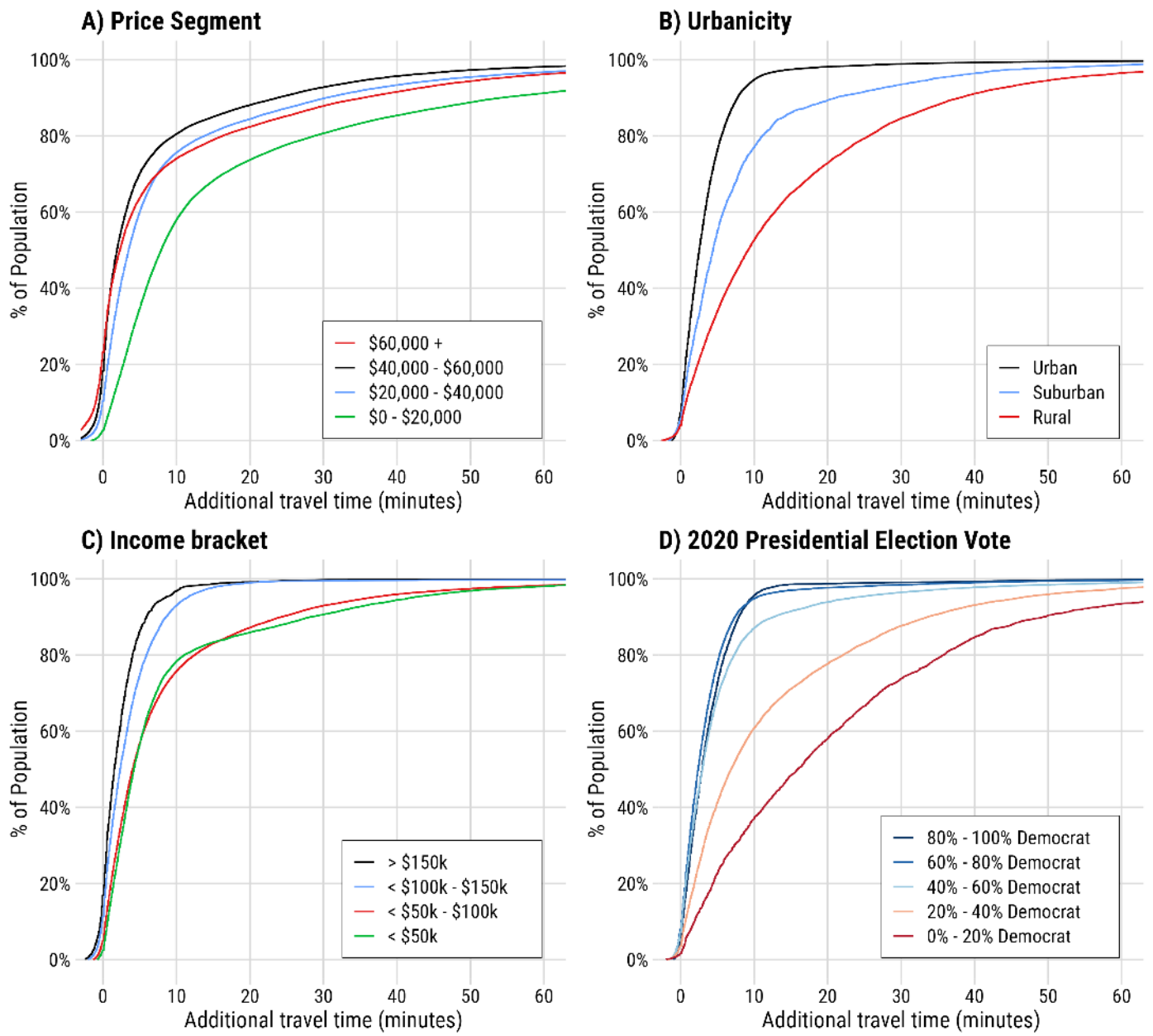
*Road travel times obtained using Open Source Routing Machine (OSRM)



PEV travel times are converging towards conventional vehicle times

80% of pop:

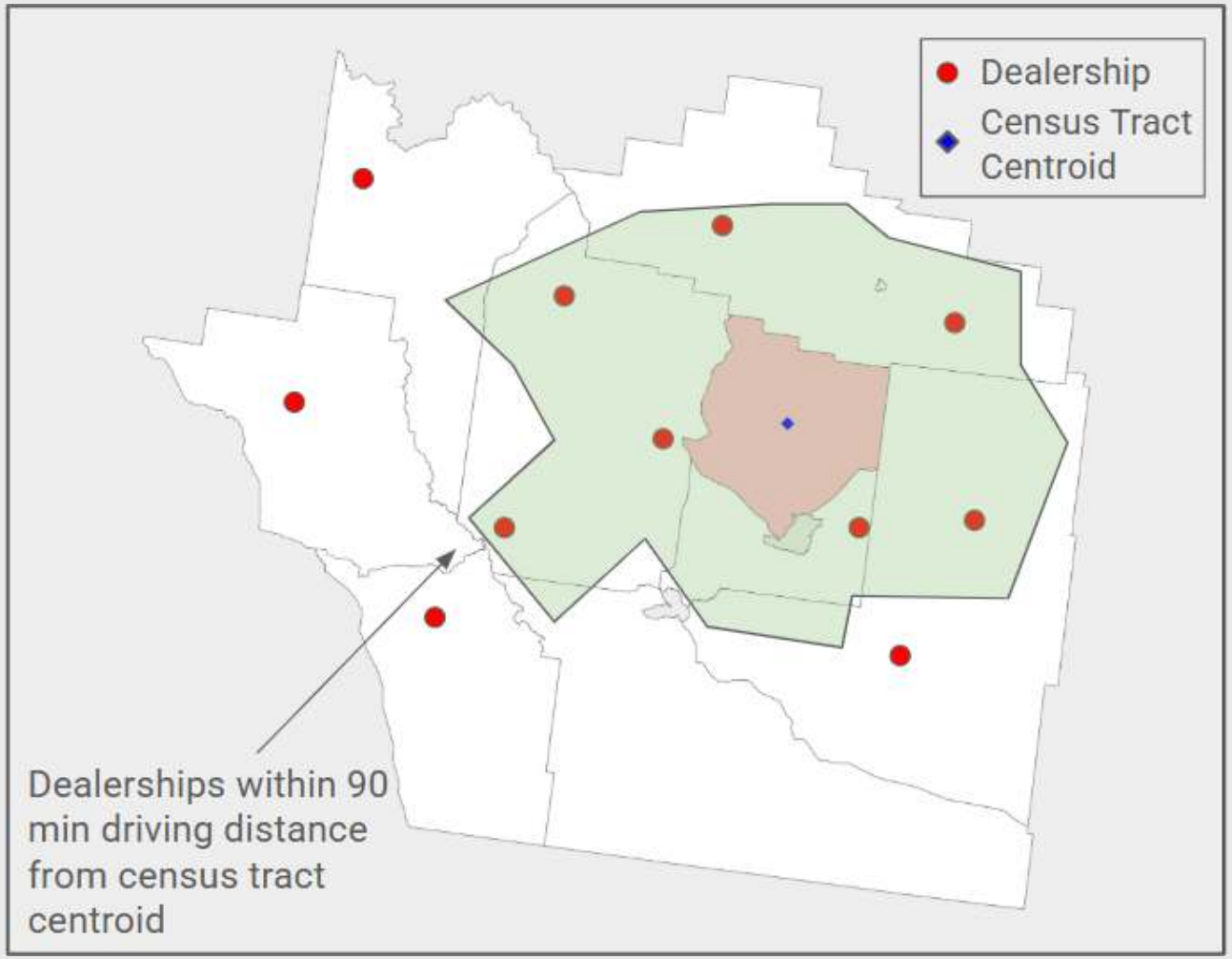
- CV in ~12 min
- PEV in ~22 min (2024)
- PEV in ~60 min (2016)



Additional travel times (PEV - CV) by demographic blocks shows disparities

Places with worse PEV access:

- Lower income areas
- Rural areas
- Republican strongholds

Local vehicle market: Vehicles in 90 minute driving isochrone

Combine ML algorithm of predicted household vehicle budget (using Census data) with local market characteristics